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HDOC Day-36 — String Operations Using Functions

Algorithm

1. Start.
2. Declare a character array to store the string.
3. Read a string from the user using `fgets()`.
4. Remove the newline character, if present.
5. Display the menu:
   - Count characters.
   - Print each character on a new line.
   - Count vowels and consonants.
   - Exit.
6. Read the user's choice.
7. Use `switch` to perform the selected operation using a separate function.
8. For **Count Characters**:
   - Initialize `count = 0`.
   - Traverse the string until `'\0'`.
   - Increment `count` for every character.
   - Display the count.
9. For **Print Characters**:
   - Traverse the string until `'\0'`.
   - Print each character on a separate line.
10. For **Vowels and Consonants**:
    - Traverse each character.
    - Check whether it is an alphabet.
    - If it is `a, e, i, o, u` in either case, increment vowel count.
    - Otherwise, increment consonant count.
    - Ignore spaces, digits, and special characters.
11. Repeat the menu until the user selects Exit.
12. Stop.

Test Case Table

Test Case	Input String	Operation	Expected Output
1	Hello	Count characters	
2	ABC	Print characters	A, B, C on separate lines
3	Education	Count vowels/ consonants	Vowels = 5, Consonants = 4

4 `Hello World` Count characters `11` including the space
5 `C`
`Programming`
Count vowels/
consonants
Vowels = `3`, Consonants =
`9`
6 `123 @#` Count vowels/
consonants
Vowels = `0`, Consonants =
`0`
7 Any string Exit Program terminates

C Program

```
#include <stdio.h>
int countCharacters(char str[])
{ int count = 0;
while (str[count] != '\0')
{
count++;
}
return count;
void printCharacters(char str[])
}
{
int i = 0;
printf("\nCharacters in the string:\n");
while (str[i] != '\0')
{
printf("%c\n", str[i]);
i++;
}
```

```
}
void countVowelsConsonants(char str[])
{
int i = 0;
int vowels = 0, consonants = 0;
char ch;
while (str[i] != '\0')
{
ch = str[i];
if ((ch >= 'A' && ch <= 'Z') ||
(ch >= 'a' && ch <= 'z'))
{
if (ch == 'A' || ch == 'E' || ch == 'I' ||
ch == 'O' || ch == 'U' ||
ch == 'a' || ch == 'e' || ch == 'i' ||
ch == 'o' || ch == 'u')
{
vowels++;
}
else
{
consonants++;
}
}
```

```
i++;
}
printf("Number of vowels = %d\n", vowels);
printf("Number of consonants = %d\n", consonants);
}
int main()
{
char str[200];
int choice;
int i;
printf("Enter a string: ");
fgets(str, sizeof(str), stdin);
/* Remove newline added by fgets */
i = 0;
while (str[i] != '\0')
{
if (str[i] == '\n')
{
str[i] = '\0';
break;
}
i++;
}
```

```
do
{
printf("\n----- MENU -----\n");
printf("1. Count characters in the string\n");
printf("2. Print each character on a new line\n");
printf("3. Count vowels and consonants\n");
printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch (choice)
{
case 1:
printf("Number of characters = %d\n",
countCharacters(str));
break;
case 2:
printCharacters(str);
break;
case 3:
countVowelsConsonants(str);
break;
case 4:
printf("Exiting the program...\n");
```

```
break;
default:
printf("Invalid choice. Please enter 1 to 4.\n");
}
} while (choice != 4);
return 0;
}
```